

More about the calculator

- The complete syntax descripton.
- Syntax errors.
- How to start.



Demonstration

Input : $4 - (5 - 2 * 2) / (1 + 1) - (3 - 1)$

Result: 1.5

Input : $1 + 2 + 3 + 4 = x$

Result: 10.0

Input : $(2 * x = y = z) - x / 2 - x$

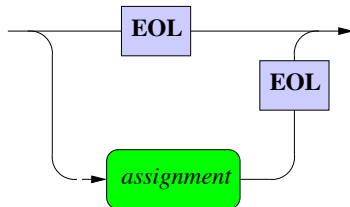
Result: 5.0



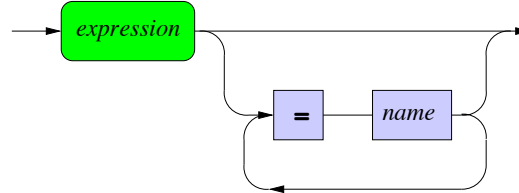


Syntax charts

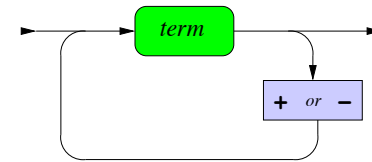
statement



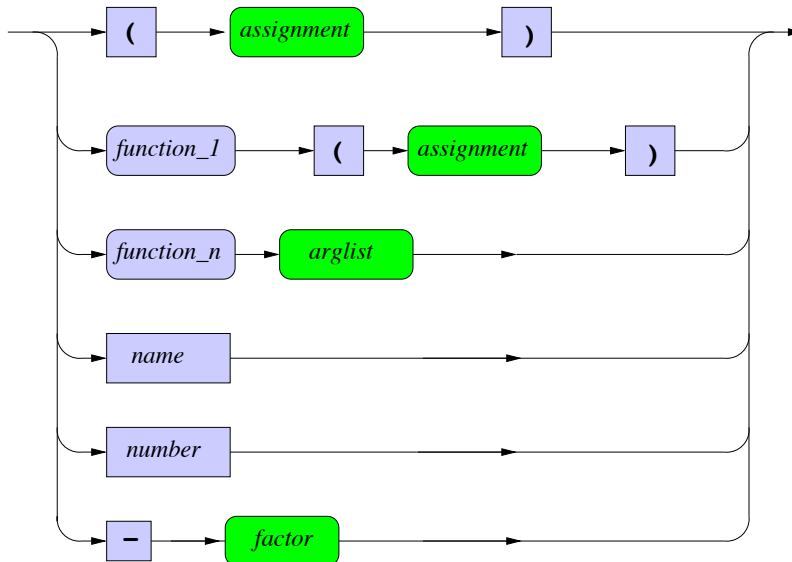
assignment



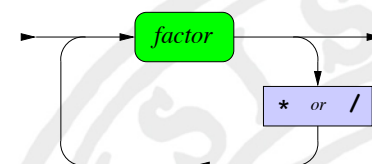
expression



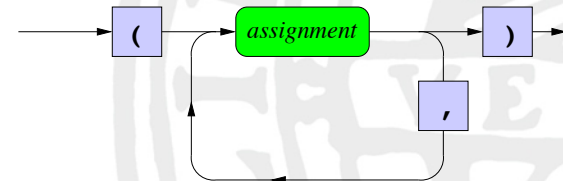
factor



term



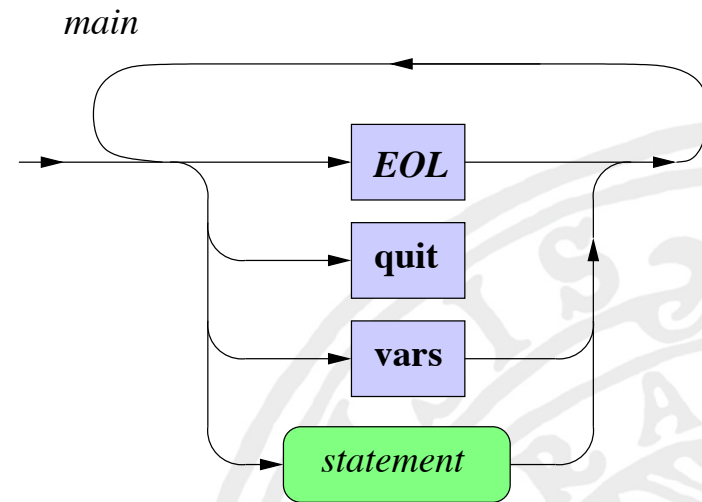
arglist



A main-function

We also need a function that drives it all :

This function starts by reading lines one from a file that is attached.
The idea is that you should fill the file with your own tests.



Syntax error in the input

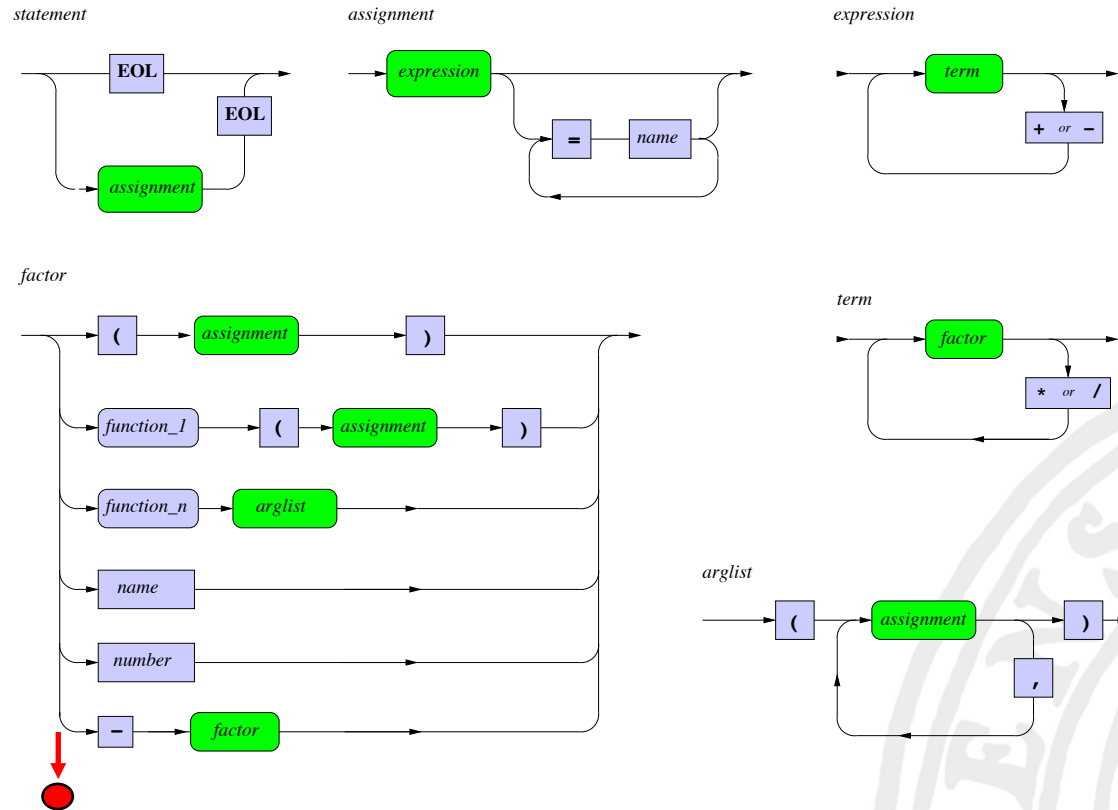
What happens if the user types an expression that is not syntactically correct?

Example:

- $1 + * y$
- $+2 + 3$
- $4 = 5$
- $(1=x+1)$
- $1) 2$
- $(2$



Input: 1 + * y



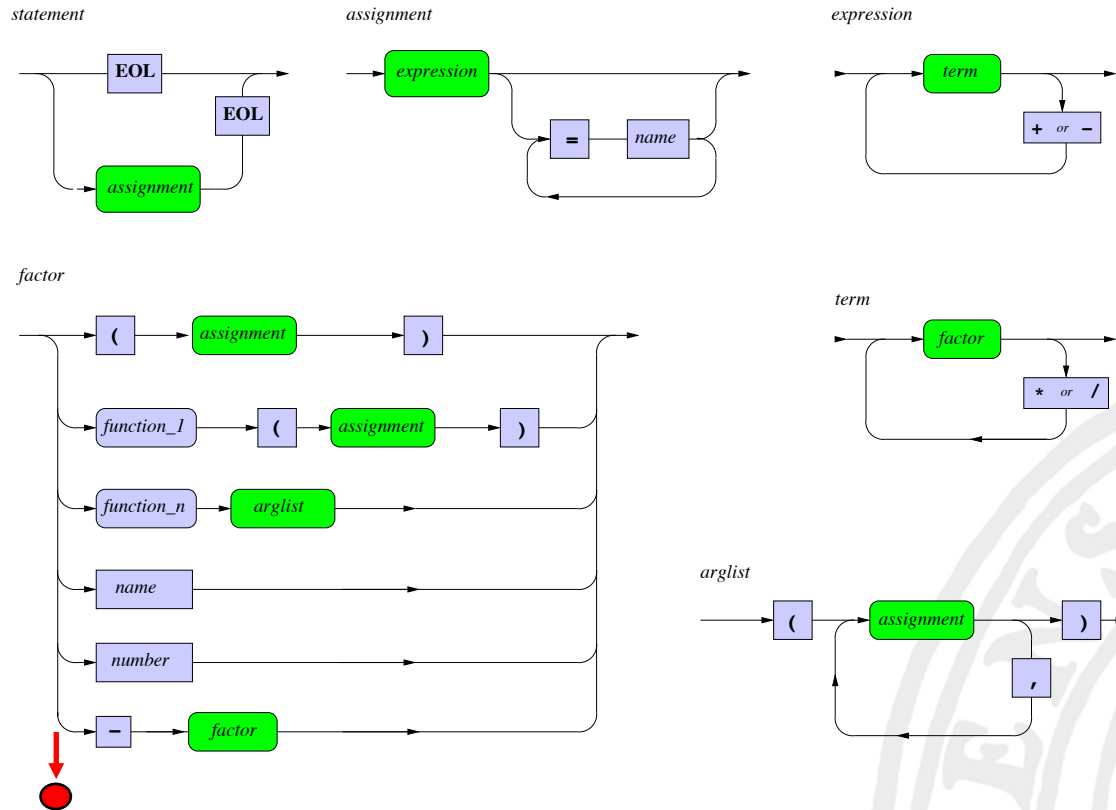
Input: 1 + * y

*** Syntax error: Expected number, word or '('

Error occurred at '*' just after '+'

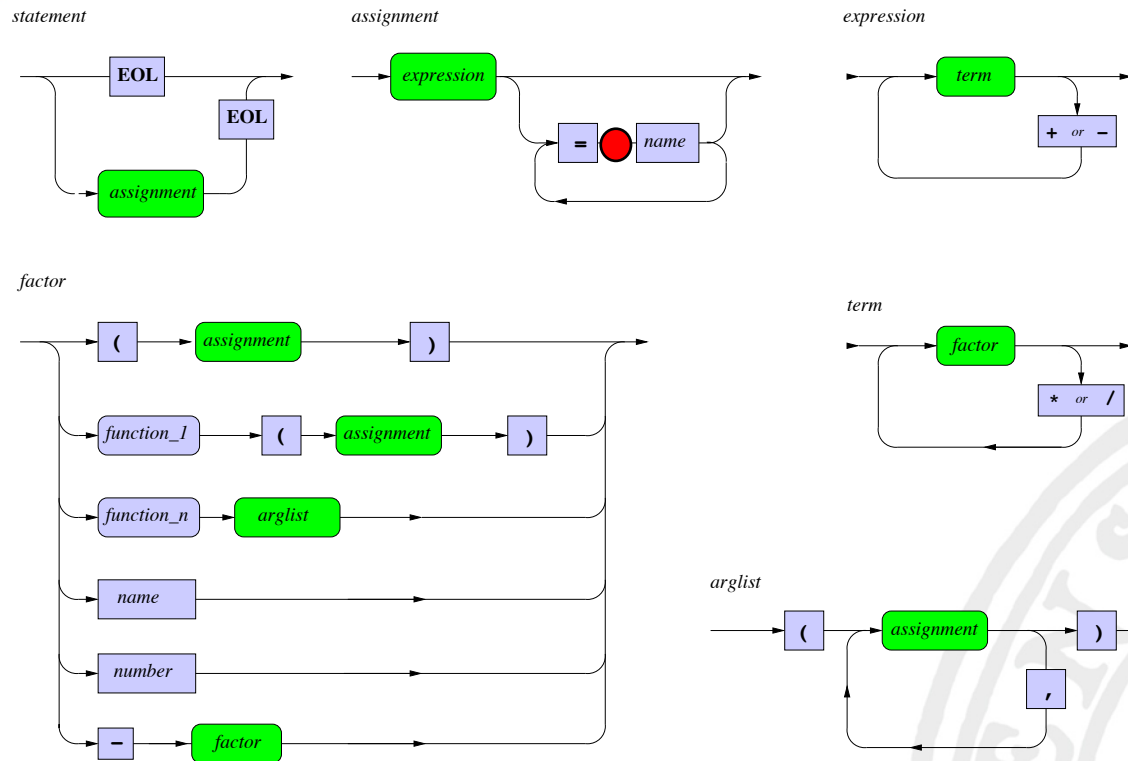


Input: +2 + 3



Input : +2+3

*** Syntax error: Expected number, word or '('
Error occurred at '+' just after 'START'

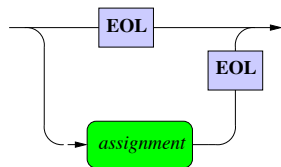


```
*** Syntax error: Expected name after '='
Error occurred at '5' just after '='
```

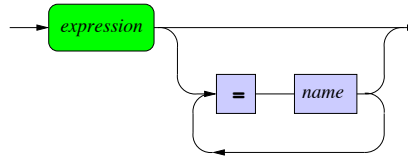


Input: (1 = x + 1)

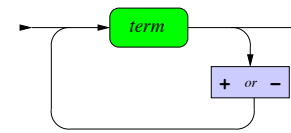
statement



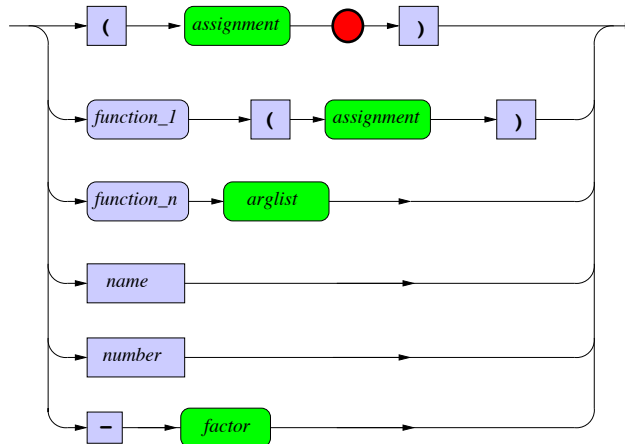
assignment



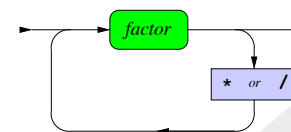
expression



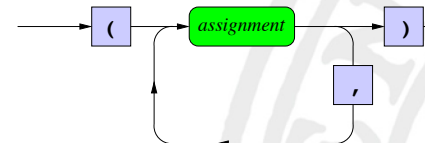
factor



term



arglist



Input : (1 = x + 1)

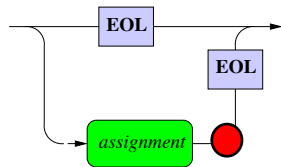
*** Syntax error: Expected ')'

Error occurred at '+' just after 'x'

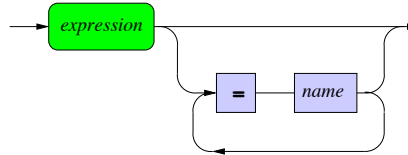


Input: 1) 2

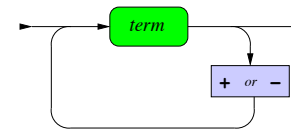
statement



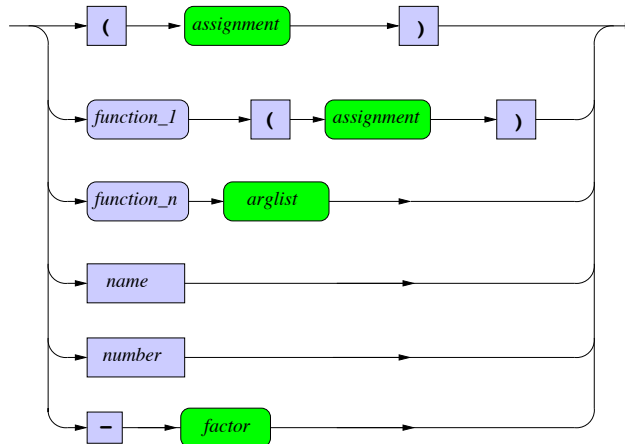
assignment



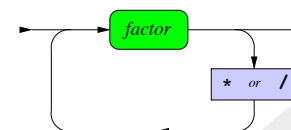
expression



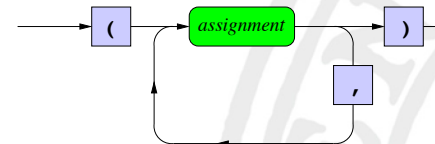
factor



term



arglist



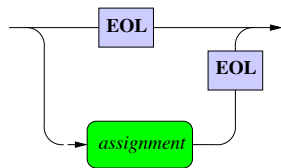
Input : 1) 2

*** Syntax error: Expected end of line
Error occurred at ')' just after '1'

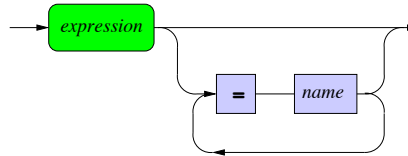


Input: (2

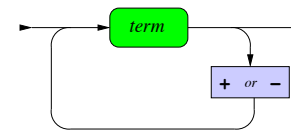
statement



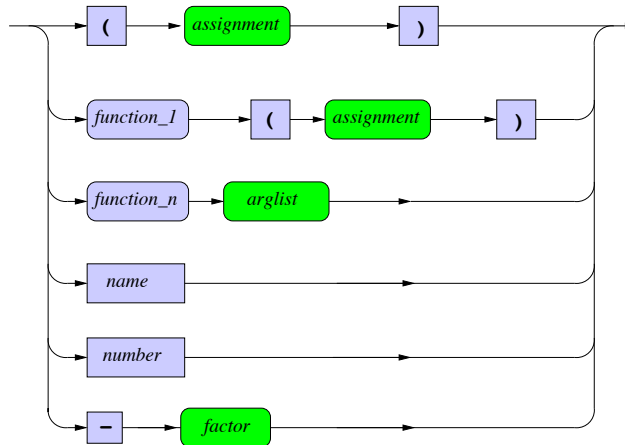
assignment



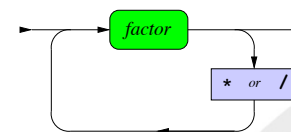
expression



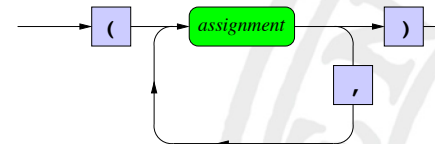
factor



term



arglist



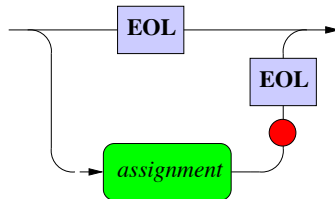
Input : (2

*** Syntax error: Unbalanced parentheses

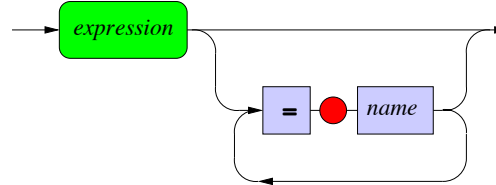


Ställen att kontrollera

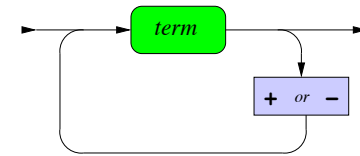
statement



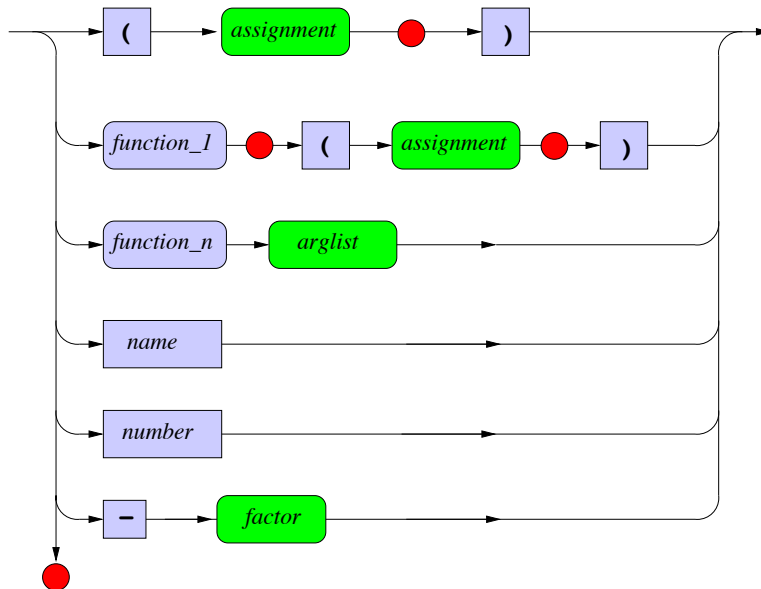
assignment



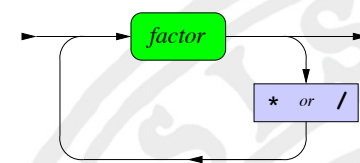
expression



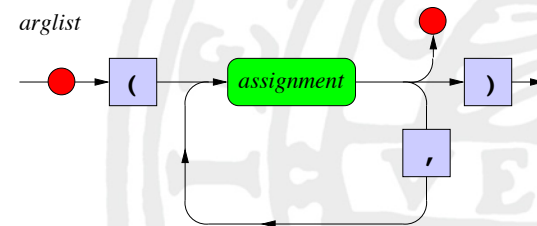
factor



term



arglist



How do you get started?

- Run the downloaded program! It should work directly for addition, multiplication and parentheses.
- Add subtraction and division. Make sure expressions like $1+2-3+4$ work!
- Addn unary minus ie so that eg $-(3+2)$ works.
- Write the class `EvaluationError`. Make sure it's used when dividing by 0. These exceptions should be caught caught in *main*.
- Add variable handling ie give variable values in *assignment* and look up variable values in *factor*. Undefined variables should raise evaluation `EvaluationError`.
- Have accurate error messages about what was expected.
- Do *not* proceed until you get these things working!



UPPSALA
UNIVERSITET

The end

